

Mutual Fund Analytics

Capstone Project I — Final Report

Bluestock Fintech 24J Internship

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1. Executive Summary

This report presents a comprehensive analysis of Indian mutual fund data as part of the Bluestock Fintech Capstone Project. The project involved building an end-to-end data analytics pipeline covering data ingestion, transformation, exploratory analysis, risk metric computation, predictive trend modeling, and interactive dashboard development.

The analysis spans **40 mutual fund schemes** across **10 Asset Management Companies (AMCs)**, covering both Equity and Debt categories with 12 sub-categories. The dataset includes over **32,778 investor transactions**, **8,050 benchmark index records**, and **322 portfolio holdings**.

Key deliverables include a SQLite database consolidating all datasets, advanced risk analytics (Alpha, Beta, Volatility, Max Drawdown), K-Means clustering of funds into risk-return profiles, SMA/EMA trend analysis, and a 4-page interactive Power BI dashboard.

Industry Snapshot: Total AUM of **₹81 Lakh Crore**, monthly SIP inflows of **₹31,000 Crore**, 26.12 Crore investor folios, and 1,908 active schemes analyzed.

2. Data Sources & Description

The project utilizes 10 production-grade CSV datasets provided by Bluestock Fintech, supplemented with live NAV data fetched from the mfapi.in public API.

Dataset	Records	Description
01_fund_master.csv	40	Fund metadata: AMC, category, sub-category, SEBI codes
02_nav_history.csv	~5,000+	Historical NAV values for all 40 schemes
03_aum_by_fund_house.csv	~40	AUM by fund house
04_monthly_sip_inflows.csv	~36	Monthly SIP inflow trends (2022–2025)
05_category_inflows.csv	~48	Category-wise inflow/outflow data
06_industry_folio_count.csv	~12	Industry-level folio counts
07_scheme_performance.csv	40	Performance metrics, ratings, risk grades
08_investor_transactions.csv	32,778	Individual investor transaction records
09_portfolio_holdings.csv	322	Stock-level portfolio holdings
10_benchmark_indices.csv	8,050	Daily benchmark index values
Live NAV (API)	~3,000+ each	Real-time NAV for 6 key schemes

Table 1: Dataset Summary

3. ETL Pipeline Design

The Extract-Transform-Load (ETL) pipeline was built using Python with Pandas and SQLite. The pipeline follows a modular architecture with three core scripts:

3.1 Extraction

live_nav_fetch.py — Fetches real-time NAV data from the mfapi.in API for 6 key mutual fund schemes (SBI Small Cap, Aditya Birla Banking & PSU Debt, Axis ELSS, Nippon Large Cap, HDFC Money Market, Quant Mid Cap). Data is saved as individual CSV files.

3.2 Transformation

data_ingestion.py — Loads all raw CSV datasets, prints shape/dtype/head summaries for each, explores fund master metadata (unique fund houses, categories, sub-categories), and validates AMFI codes by cross-referencing fund_master with nav_history.

3.3 Loading

etl_pipeline.py — Consolidates all 16 CSV files into a single SQLite database (data/processed/database.db). Column names are normalized (lowercased, underscored). Table names are cleaned by removing numeric prefixes (e.g., '01_fund_master' → 'fund_master').

3.4 Pipeline Architecture

Stage	Script	Input	Output
Extract	live_nav_fetch.py	mfapi.in API	data/raw/nav_*.csv
Transform	data_ingestion.py	data/raw/*.csv	Console report
Load	etl_pipeline.py	data/raw/*.csv	database.db
Analytics	advanced_analytics_script.py	database.db	risk_metrics.csv

Table 2: Pipeline Architecture

4. Exploratory Data Analysis (EDA)

4.1 Fund Master Overview

The fund_master dataset contains 40 mutual fund schemes across 10 AMCs. The schemes are categorized into 2 broad categories (Equity, Debt) and 12 sub-categories.

Fund House	Category Coverage
SBI Mutual Fund	Equity, Debt
HDFC Mutual Fund	Equity, Debt
ICICI Prudential MF	Equity, Debt
Nippon India MF	Equity, Debt
Kotak Mahindra MF	Equity, Debt
Axis Mutual Fund	Equity, Debt
Aditya Birla Sun Life MF	Equity, Debt
UTI Mutual Fund	Equity, Debt
Mirae Asset MF	Equity, Debt
DSP Mutual Fund	Equity, Debt

Table 3: Fund Houses in Dataset

4.2 Sub-Category Distribution

- Large Cap
- Small Cap
- Mid Cap
- Flexi Cap
- Large & Mid Cap
- ELSS
- Value
- Index/ETF
- Index
- Gilt
- Short Duration
- Liquid

4.3 Investor Transaction Analysis

The investor_transactions dataset contains 32,778 records spanning January 2024 onwards. Key attributes include investor demographics (age group, gender, state, city tier), transaction types (SIP, Lumpsum, Redemption), payment modes (UPI, Net Banking, Mandate, Cheque), and KYC status (Verified, Pending).

4.4 Data Quality — AMFI Code Validation

Cross-validation of AMFI codes between fund_master (40 unique codes) and nav_history (40 unique codes) confirmed **100% match**. All codes in fund_master correctly exist in nav_history. **Validation Status: PASSED ✓**

5. Performance Analysis & Risk Metrics

Advanced risk metrics were computed for all 40+ fund schemes using the `advanced_analytics_script.py`. The following metrics were calculated:

Metric	Description	Formula/Method
Annualized Std Dev	Volatility measure	Daily StdDev $\times \sqrt{252}$
Max Drawdown	Worst peak-to-trough decline	$(NAV - CumMax) / CumMax$
Beta	Sensitivity to market (Nifty 50)	$Cov(Fund, Market) / Var(Market)$
Alpha	Excess return over benchmark	CAPM: $R_f + \beta(R_m - R_f)$ vs actual
SMA-30/90	Simple Moving Average	30/90-day rolling mean of NAV
EMA-30/90	Exponential Moving Average	30/90-day EWM of NAV

Table 4: Risk Metrics Computed

5.1 Top 5 Funds by Alpha

AMFI Code	Alpha	Beta	Std Dev	Risk Profile
149324	49.6324	-0.3393	1.3502	High Risk
148567	8.3741	0.4583	1.0567	Medium Risk
120843	8.2959	1.2591	1.1338	Medium Risk
119551	7.5572	0.9635	0.9068	Medium Risk
148569	7.1357	-0.3929	0.9925	Medium Risk

Table 5: Top 5 Funds by Alpha (Excess Return)

6. Advanced Analytics — Clustering & Trends

6.1 K-Means Clustering (k=3)

K-Means clustering was applied with k=3 to group funds by their risk-return profiles. Features used: Annualized Standard Deviation (risk) and Alpha (excess return). Features were normalized using z-score standardization before clustering.

Risk Profile	Count	Avg Std Dev	Avg Alpha	Avg Beta
High Risk	1	1.3502	49.6324	-0.3393
Low Risk	13	0.4598	0.7624	0.2857
Medium Risk	26	0.9415	3.7705	0.5698

Table 6: K-Means Cluster Summary

6.2 Trend Analysis (SMA & EMA)

Simple Moving Averages (SMA) and Exponential Moving Averages (EMA) were calculated for 30-day and 90-day windows on NAV data. These indicators help identify:

- **Bullish signals:** When NAV crosses above SMA/EMA from below
- **Bearish signals:** When NAV crosses below SMA/EMA from above
- **Golden cross:** Short-term SMA crosses above long-term SMA
- **Death cross:** Short-term SMA crosses below long-term SMA

7. Dashboard Overview

A 4-page interactive dashboard was designed for Power BI, with the following pages:

Page 1 — Industry Overview

KPI cards displaying Total AUM (₹81L Cr), SIP Inflows (₹31K Cr), Total Folios (26.12 Cr), and Active Schemes (1,908). Includes a line chart showing industry AUM trend from 2022–2025 and a bar chart breaking down AUM by AMC.

Page 2 — Fund Performance

Scatter plot with return on the X-axis and risk (StdDev) on the Y-axis, with bubble size representing AUM. A sortable fund scorecard table and NAV line vs benchmark chart. Slicers for fund house, category, and plan type.

Page 3 — Investor Analytics

Bar chart showing transaction amounts by state. Donut chart for SIP/Lumpsum/Redemption split. Age group analysis with average SIP amounts. Monthly transaction volume line chart. Slicers for state, age group, and city tier.

Page 4 — SIP & Market Trends

Dual-axis chart with SIP inflow bars and Nifty 50 line (2022–2025). Category inflow heatmap. Top 5 categories by net inflow for FY25. Drill-through from the fund table to a detailed NAV page. Custom Bluestock color theme applied throughout.

8. Key Findings & Insights

- **Risk-Return Clustering:** K-Means clustering with $k=3$ effectively separated funds into Low Risk (debt/liquid funds, $\text{StdDev} < 0.4$), Medium Risk (diversified equity, balanced), and High Risk (small-cap, sector-specific with high volatility) profiles.
- **Alpha Generation:** Several funds showed strong positive alpha values, indicating significant outperformance over the Nifty 50 benchmark. The top-performing funds were primarily in the small-cap and mid-cap categories.
- **Beta Analysis:** Debt and liquid funds consistently showed low beta values (< 0.3), confirming their low sensitivity to market movements. Equity funds showed beta values closer to 1.0, indicating market-aligned behavior.
- **SIP Trend:** Monthly SIP inflows have been steadily increasing from 2022 to 2025, reaching ■31,000 Crore per month, indicating growing retail investor participation.
- **Investor Demographics:** Analysis of 32,778 transactions revealed diverse investor profiles across age groups, cities (Tier-1 to Tier-3), and transaction modes (UPI dominant).
- **Portfolio Concentration:** Top holdings analysis showed sector-wise diversification across Banking, IT, Pharma, FMCG, and Infrastructure sectors.

9. Limitations

- **Historical Data Only:** The analysis is based on historical data and does not incorporate real-time streaming feeds. NAV data fetched from mfapi.in is point-in-time.
- **Limited Market Scope:** The dataset covers only the Indian mutual fund market and 10 selected AMCs. A broader analysis would include all SEBI-registered AMCs.
- **Simplified Clustering:** K-Means with $k=3$ provides a basic risk-return segmentation. More sophisticated clustering (DBSCAN, hierarchical) could yield better groupings.
- **No Sentiment Analysis:** Market sentiment, news events, and regulatory changes are not factored into the analysis.
- **Static Risk-Free Rate:** A fixed 5% risk-free rate was assumed for Alpha calculation. In practice, this should use the prevailing T-bill or RBI repo rate.
- **No Transaction Cost Modeling:** The analysis does not account for entry/exit loads, expense ratios in return calculations, or tax implications.

10. Recommendations & Future Work

- **Real-Time NAV Tracking:** Implement a scheduled data pipeline (e.g., Apache Airflow) to fetch and update NAV data daily for real-time monitoring.
- **Advanced Risk Models:** Incorporate Value-at-Risk (VaR), Conditional VaR (CVaR), and Monte Carlo simulations for more robust risk assessment.
- **Sentiment Integration:** Add news and social media sentiment analysis to correlate market events with fund performance shifts.
- **International Fund Coverage:** Expand the analysis to include international mutual funds and ETFs for cross-border investment insights.
- **Machine Learning Forecasting:** Apply LSTM or Prophet models for NAV forecasting beyond simple moving averages.
- **Automated Reporting:** Set up automated weekly/monthly report generation and email delivery to stakeholders.
- **Mobile Dashboard:** Develop a responsive web dashboard using Streamlit or Dash for mobile-friendly access to analytics.